

Milan Jani

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CAREER OBJECTIVE

ICT undergraduate focused on VLSI, digital design, and Verilog HDL, with embedded systems experience on ARM, ESP32, and MSP430. Patent holder seeking opportunities in VLSI and embedded systems.

EDUCATION

Marwadi University, B.Tech in Information & Communication Technology [July'23 - May'27]

- GPA: 9.03/10
- **Coursework:** Embedded Systems, Digital Electronics, Advanced Microprocessors, Computer Architecture, Data Structures & Algorithms, Computer Networks

EXPERIENCE

VLSI Intern | Microcircuits Innovations Pvt. Ltd., Ahmedabad [June'25 - July'25]

- Automated design flow tasks using **50+ Linux commands**, Shell/TCL scripting, and Makefile flow.
- Explored ASIC flow, circuit layout, and backend design verification using industry EDA tools.

Research Intern | ICT Department, Marwadi University, Rajkot [July'24 – Nov'25]

- Contributed to MIMO antenna research using FEM simulation for 5G/satellite communication.
- Evaluated return loss, gain, and bandwidth parameters to support antenna performance optimization.

PUBLICATIONS

LPG Cylinder Monitoring and Leak Alert System | Patent Publication [June'24]

Patent No.: 202421033238 Status: Published

PROJECTS

Smart Vehicle Entry-Exit Logging System | Link [Jan'26]

- Engineered a vehicle management system with real-time **ANPR**, **ID card OCR** (Aadhaar, DL) via **Azure Computer Vision**, and kiosk-based visitor registration deployed live on **Raspberry Pi** via **Docker**.
- Implemented **admin dashboard** with live tracking and control over visitor and vehicle records.

Robotic Arm Development Project | Project Contributor | Link [Oct'25 - March'26]

- Applied robot control concepts using **NVIDIA Isaac Sim** and **Isaac Lab** for simulation and **RL training**.
- Contributed to robotic arm design and testing by integrating motion planning and control algorithms.

Adaptive Embedded Security and Automation System | Link [Oct'25]

- Designed an interrupt-driven security system on **MSP430F5529** with **3-level hardware priority** (fire > SOS > gas/motion) and ISR-based flag architecture handling 5 concurrent sensor threats.
- Configured **UART**, **I2C**, **ADC**, and **PWM** peripherals with real-time LCD feedback and <1ms response.

LPG Cylinder Monitoring and Leak Alert System [Aug'23]

(Guides: Prof. C.D. Parmar, Prof. Vijay Dubey) Course Project

- Built an IoT-based LPG system that tracks cylinder weight and sends alerts **2–3 days** early for booking.
- Implemented gas leak sensors and responsive LED alerts providing **24×7** real-time safety notifications.

SKILLS & TECHNOLOGIES

Languages	MCUs / SBCs	Tools & Platforms	Protocols
C, C++, Embedded C, Python, Verilog (learning)	ARM, AVR, MSP430, ESP32, ESP8266, Raspberry Pi	Keil, CCS, Arduino IDE, KiCad, LTspice, Microwind, Proteus, Git	UART, SPI, I2C, CAN

CERTIFICATIONS & PARTICIPATIONS

Digital Design using Verilog – NPTEL Online Certification [Jan'26 - Apr'26]

Semiconductor Fabrication and Manufacturing Overview – Industry Workshop [Feb'26]